GOMS/KLM Process

Admin sending message to customer (Zac, path0171)

Assumptions:

- User is an **admin**
- The user is using a laptop or a desktop
- User is using keyboard and mouse or trackpad
- The user has no physical disadvantage that would affect the process
- System response time is minimal (0.5-1s, included as R)
- User is an experienced admin (has done this action many times before)
- User is a retired volunteer (non-skilled typist)
- User is already on the homepage before starting the process

GOMS Process:

- Goal - The state the user wants to achieve
- Methods - The cognitive processes needed to attain the goals
- Operators - The procedures of methods for accomplishing goals
- Selection Rules - Decide which method to select when there is more than one

Goal:

- The user (admin) wants to send a message to a customer through the admin dashboard.

Methods:

Method 1:

1. Access homepage
2. Sign in (if not already)
3. Navigate to the dashboard
4. Open Messages tab
5. Select customer
6. Type message
7. Send message

Method 2:

1. Access homepage
2. Sign in (if not already)
3. Navigate to the dashboard
4. Click “View All” in the recent messages section
5. Select customer
6. Type message
7. Send message

Method 3:

1. Access homepage
2. Sign in (if not already)
3. Navigate to the dashboard
4. Click the customer conversation in the recent messages section
5. Type message
6. Send message

Operators:

Method 1:

1. Open homepage
2. Move the mouse to the “Sign In” button (if signed in move to step 9)
3. Click “Sign In”
4. Move the cursor to the email field
5. Type email
6. Move the cursor to the password field
7. Type password
8. Click the “Sign In” button
9. Wait for the dashboard to load
10. Move the mouse to the sidebar
11. Click the “Messages” tab
12. Move the mouse over the customer list
13. Click the desired customer
14. Move the cursor to the message input field
15. Click the input field
16. Type message
17. Press Enter

Method 2:

1. Open homepage
2. Move the mouse to the “Sign In” button (if signed in move to step 9)
3. Click “Sign In”
4. Move the cursor to the email field
5. Type email
6. Move the cursor to the password field
7. Type password
8. Click the “Sign In” button
9. Wait for the dashboard to load
10. Move the mouse to the “Recent Messages” section
11. Click the “View All” button
12. Move the mouse over the customer list
13. Click the desired customer
14. Move the cursor to the message input field
15. Click the input field
16. Type message

17. Press Enter

Method 3:

1. Open homepage
2. Move the mouse to the “Sign In” button (if signed in move to step 9)
3. Click “Sign In”
4. Move the cursor to the email field
5. Type email
6. Move the cursor to the password field
7. Type password
8. Click the “Sign In” button
9. Wait for the dashboard to load
10. Move the mouse over the customer list in the “Recent Messages” section
11. Click the desired customer
12. Move the cursor to the message input field
13. Click the input field
14. Type message
15. Press Enter

Selection Rules:

- If the user is already signed in, just entering the URL for an admin sends you to the dashboard directly and not the homepage, as it avoids unnecessary steps and is faster.
- If the user is not signed in. The user must sign in with the button at the top right of the home page, as access to messaging requires authentication.
- If the message is recent enough, the desired customer might appear in the “Recent Messages” section. If this is the case, you can directly select the message, which will send you straight to the message chat. This is the fastest and most convenient option.
- If the customer is not immediately visible in the “Recent Messages” list, you can select “View All”, which will send you directly to the messages tab. A similar time is taken to select the messages tab on the left.
- If you know the customer’s message is not recent enough to be in “Recent Messages”, you can go directly to the messages by selecting the Messages tab on the left sidebar, sending you straight to the full customer list.
- If speed is important, respond quickly. The preferred method is the direct recent messages method, as it has the lowest time cost.
- If accuracy is more important, to ensure selecting the correct customer, you can use the Messages tab to view a complete and organised list of message history.

KLM Process:

Operators:

- **K** = Keystroke (0.28s)
- **P** = Pointing (1.10s)
- **P_1** = Clicking (0.20s)
- **H** = Home (0.40s)
- **M** = Mental prep (1.35s)
- **R** = System response (1.0s assumed)

Method 1 (Via messages tab):

1. Locate the "Sign In" button at the top right corner of the page (P)
2. Left-click "Sign In" (P_1)
3. Move the mouse to the email box (P)
4. Left-click the box (P_1)
5. Move hands to keyboard (H)
6. Think about what to type (M)
7. Enter email address (K)
8. Move hands to the mouse (H)
9. Move the mouse to the password box (P)
10. Left-click the box (P_1)
11. Move hands to keyboard (H)
12. Think about what to type (M)
13. Enter password (K)
14. Press enter to submit (K)
15. Move hands to the mouse (H)
16. Locate the sidebar on the left side of the screen (P)
17. Left-click "Messages" tab (P_1)
18. Think about the customer you need to message (M)
19. Scan the customer message list and move the mouse to the desired customer (P)
20. Select the desired customer to expand chat (P_1)
21. Move the mouse to the message input field (P)
22. Left-click the input field (P_1)
23. Move hands to keyboard (H)
24. Think about what message to type (M)
25. Type desired message (K)
26. Click the send button to the right of the input field or press Enter (K)
27. Wait for message to send (R)

Condensed:
E = Letters in Email (Approx 20 Characters)
W = Letters in Password (Approx 8 Characters)
M = Letters in Message (Approx 5-25 Characters)

- **K** = Keystroke (0.28s)
- **P** = Pointing (1.10s)
- **P_1** = Clicking (0.20s)

- **H** = Home (0.40s)
- **M** = Mental prep (1.35s)
- **R** = System response (1.0s assumed)

PP_1 PP_1 HMK(E) HPP_1 HMK(W)K HPP_1 MPP_1 PP_1 HMK(M)KR

Cost:
5 character message: 1.1 + 0.2 + 1.1 + 0.2 + 0.4 + 1.35 + (0.28 x 20) + 0.4 + 1.1 +0.2 + 0.4 +1.35 + (0.28 x 8) + 0.28 + 0.4 + 1.1 + 0.2 + 1.35 + 1.1 + 0.2 + 1.1 + 0.2 + 0.4 + 1.35 + (0.28x5) + 0.28 + 1 = **26 seconds**

25 character message: 1.1 + 0.2 + 1.1 + 0.2 + 0.4 + 1.35 + (0.28 x 20) + 0.4 + 1.1 +0.2 + 0.4 +1.35 + (0.28 x 8) + 0.28 + 0.4 + 1.1 + 0.2 + 1.35 + 1.1 + 0.2 + 1.1 + 0.2 + 0.4 + 1.35 + (0.28x25) + 0.28 + 1 = **31.6 seconds**

Method 2 (Via recent messages - "View All"):

1. Locate the "Sign In" button at the top right corner of the page (P)
2. Left-click "Sign In" (P_1)
3. Move the mouse to the email box (P)
4. Left-click the box (P_1)
5. Move hands to keyboard (H)
6. Think about what to type (M)
7. Enter email address (K)
8. Move hands to the mouse (H)
9. Move the mouse to the password box (P)
10. Left-click the box (P_1)
11. Move hands to keyboard (H)
12. Think about what to type (M)
13. Enter password (K)
14. Press enter to submit (K)
15. Move hands to the mouse (H)
16. Move the mouse to the "Recent Messages" section on the dashboard (P)
17. Left-click "View All" button (P_1)
18. Think about the customer you need to message (M)
19. Scan the customer message list and move the mouse to the desired customer (P)
20. Select the desired customer to expand chat (P_1)
21. Move the mouse to the message input field (P)
22. Left-click the input field (P_1)
23. Move hands to keyboard (H)
24. Think about what message to type (M)
25. Type desired message (K)
26. Click the send button to the right of the input field or press Enter (K)
27. Wait for message to send (R)

Condensed:
E = Letters in Email (Approx 20 Characters)
W = Letters in Password (Approx 8 Characters)
M = Letters in Message (Approx 5-25 Characters)

PP_1 PP_1 HMK(E) HPP_1 HMK(W)K HPP_1 MPP_1 PP_1 HMK(M)KR

Cost:
5 character message: 1.1 + 0.2 + 1.1 + 0.2 + 0.4 + 1.35 + (0.28 x 20) + 0.4 + 1.1 +0.2 + 0.4 +1.35 + (0.28 x 8) + 0.28 + 0.4 + 1.1 + 0.2 + 1.35 + 1.1 + 0.2 + 1.1 + 0.2 + 0.4 + 1.35 + (0.28x5) + 0.28 + 1 = **26 seconds**

25 character message: 1.1 + 0.2 + 1.1 + 0.2 + 0.4 + 1.35 + (0.28 x 20) + 0.4 + 1.1 +0.2 + 0.4 +1.35 + (0.28 x 8) + 0.28 + 0.4 + 1.1 + 0.2 + 1.35 + 1.1 + 0.2 + 1.1 + 0.2 + 0.4 + 1.35 + (0.28x25) + 0.28 + 1 = **31.6 seconds**

Method 3 (Via recent messages directly):

1. Locate the "Sign In" button at the top right corner of the page (P)
2. Left-click "Sign In" (P_1)
3. Move the mouse to the email box (P)
4. Left-click the box (P_1)
5. Move hands to keyboard (H)
6. Think about what to type (M)
7. Enter email address (K)
8. Move hands to the mouse (H)
9. Move the mouse to the password box (P)
10. Left-click the box (P_1)
11. Move hands to keyboard (H)
12. Think about what to type (M)
13. Enter password (K)
14. Press enter to submit (K)
15. Move hands to the mouse (H)
16. Think about the customer you need to message (M)
17. Move the mouse to the "Recent Messages" section on the dashboard (P)
1. Left-click the desired customer from the list to expand chat (P_1)
18. Move the mouse to the message input field (P)
19. Left-click the input field (P_1)
20. Move hands to keyboard (H)
21. Think about what message to type (M)
22. Type desired message (K)
23. Click the send button to the right of the input field or press Enter (K)
24. Wait for message to send (R)

Condensed:
E = Letters in Email (Approx 20 Characters)
W = Letters in Password (Approx 8 Characters)
M = Letters in Message (Approx 5-25 Characters)

PP_1 PP_1 HMK(E) HPP_1 HMK(W)K $HMPP_1$ PP_1 HMK(M)KR

Cost:
5 character message: 1.1 + 0.2 + 1.1 + 0.2 + 0.4 + 1.35 + (0.28 x 20) + 0.4 + 1.1 +0.2 + 0.4 +1.35 + (0.28 x 8) + 0.28 + 0.4 + 1.35 + 1.1 + 0.2 + 1.1 + 0.2 + 0.4 + 1.35 + (0.28x5) + 0.28 + 1 = **24.7 seconds**

25 character message: 1.1 + 0.2 + 1.1 + 0.2 + 0.4 + 1.35 + (0.28 x 20) + 0.4 + 1.1 +0.2 + 0.4 +1.35 + (0.28 x 8) + 0.28 + 0.4 + 1.35 + 1.1 + 0.2 + 1.1 + 0.2 + 0.4 + 1.35 + (0.28x25) + 0.28 + 1 = **30.3 seconds**

Best Method:

- Method 1: 26 - 31.6 seconds
- Method 2: 26 - 31.6 seconds
- **Method 3: 24.7 - 30.3 seconds**

Conclusion:

Based on the GOMS and KLM analysis, the best solution is the direct access method via Recent Messages (Method 3), with the lowest time of (24.7 - 30.3 seconds). This indicates, in relation to efficiency and usability, that the system should prioritise quick-access features through shortcuts. Enhancing the visibility and accessibility of recent conversations will allow admins to complete tasks faster with fewer interactions, improving the overall user experience.